

```
x = 0: pi/100: pi*2;
y = sin(x);
z = cos(x);
plot(x, y, 'r');
hold on
plot(x, z, 'g');
```

The subplot() function divides the current figure into rectangular panes that are numbered row-wise. An example for 'subplot' command is given below (its output is shown in Fig. 6):

```
x = 0: pi/100: pi*2;
y = sin(x);
z = cos(x);
t = exp(-x);
u = (x.^3);
subplot(2,2,1);
plot(x,y);
subplot(2,2,2);
```

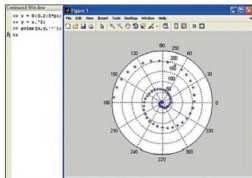


Fig. 7: Polar () function output

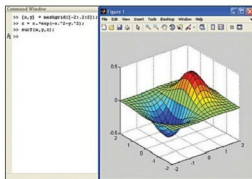


Fig. 8: 3D plotting of surface plot



Fig. 9: Reading the handel.wav file

```
plot(x,z);
subplot(2,2,3);
plot(x,t);
subplot(2,2,4);
plot(x,u);
```

An example of the polar command is given below (its output is shown in Fig. 7):

```
x = 0:0.2:5*pi;
y = x.^2;
polar(x, y, 'r');
```

Similarly, we also have other commands for 2D graphics like bar, stairs, loglog and semilog.

3D Plotting. There are different ways to plot in multiple dimensions in MATLAB. A type of 3D plot that may be useful is a surface plot, which requires you to generate some kind of x-y plane and then apply a third function as the z dimension. The example for a surface plot using surf() function is given below (its output is shown in Fig. 8):

```
[x,y] = meshgrid([-2:2:2]); %
set up 2-D plane
z = x.*exp(-x.^2-y.^2); % plot
```

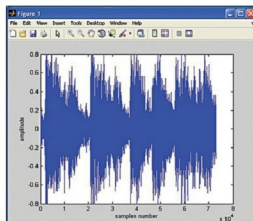


Fig. 10: The plot of handel.wav file

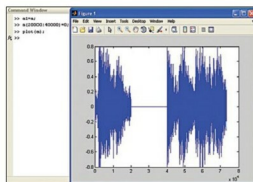


Fig. 11: Zero amplitude of samples from 20,000 to 40,000

3rd dimension on plane

```
surf(x,y,z); % surface plot
```

Similarly, we also have other commands for 3D graphics like plot3, contour and mesh.

Audio signal manipulation using MATLAB

MATLAB is a widely used environment for signal processing and analysis. Here, we introduce some key MATLAB concepts and functions that are useful to create and manipulate audio signals.

Let us use one audio sample 'wav' file and do some manipulation on it by using some commands and plot a graph of the manipulated signal.

You can download the .wav file from the Internet or create it yourself using a suitable sound recording application. Once that is done, place the .wav file in the MATLAB folder.

You can read, play and store the audio file in MATLAB. To read the file use 'wavread' command. The various syntaxes used are given below:

`y = wavread (filename)`, loads a wave file specified by the string filename, returning the sampled data in y.

`[y, Fs] = wavread (filename)`, returns the sample rate (Fs) in Hertz used to encode the data in the file.

`[y, Fs, nbits] = wavread (filename)`, returns the number of bits per sample (nbits).

For example, to read and play the handel.wav audio file, the following commands are used (its output is shown in Fig. 9):

```
a = wavread('handel1.wav'); % read the audio file.
sound(a); % to play an audio
b = flipud(a); % to reverse the audio sample
sound(b); % to play an audio
```